Calibration of Air Quality Sensors with Ensemble Stacking Random Forest, SVR, and Multiple Linear Regression

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Abstract

Acknowledgment

Table of Contents

		Page
Abstract		**iii**
Acknowledgment		**iv**
List of Tables		**viii**
List of Figures		**ix**
List of Figures		**1**
Chapter 1	**INTRODUCTION**	**1**
1.1	Background of the Study	1
1.2	Problem Statement	3
1.3	Objectives	3
1.4	Purpose of the Study	3
1.4.1	The Public	4
1.4.2	Government	4
1.4.3	Academe	4
1.4.4	ECE Field	4
1.4.5	SDG 3: Good Health	4
1.4.6	SDG 9: Innovation	4
1.4.7	SDG 11: Cities	4
1.4.8	SDG 13: Climate	5
1.4.9	SDG 17: Partners	5
1.5	Definition of Terms	5
Chapter 2	**THEORETICAL BACKGROUND**	**6**
2.1	Theories	6

2.1.1 Calibration theory 6
2.1.2 Statistical Process Control 7
2.1.3 Machine Learning 8
2.1.4 Stack Ensemble 15
2.1.5 Validation 17
2.2 Literature Review 18
2.2.1 Comparing different calibration method 18
2.2.2 Calibration using random forest 20
2.2.3 Stacked ensemble and hybrid approaches 22
Chapter 3 RESEARCH METHODOLOGY 24
3.1 Proposed Method 24
3.2 Description of Experiments 25
3.2.1 Title of Experiment of 1 25
3.2.2 Title of Experiment of 2 26
3.3 Description of Subjects 26
Chapter 4 RESULTS AND DISCUSSIONS 27
4.1 Title of Experiment of 1 27
4.2 Title of Experiment of 2 27
Chapter 5 CONCLUSIONS AND RECOMMENDATIONS 28
5.1 Conclusions 28
5.2 Recommendations 28
BIBLIOGRAPHY 29
Appendix A PRIOR ART SEARCH and ANALYSIS REPORT 30
Appendix A PATENT DRAFT 37
Appendix B TIMETABLE OF ACTIVITIES 49
Appendix C RESEARCH BUDGET 52
Appendix D ADDITIONAL FLOWCHARTS 53

Appendix E PSEUDO-CODES **54**

Appendix F ELECTRICAL CIRCUITS **55**

Appendix G SUPPLEMENTAL DATA **56**

Appendix H PLAGIARISM CHECK REPORT **57**

Appendix I ETHICS CLEARANCE **65**

Appendix J CURRICULUM VITAE **67**

List of Tables

Page

List of Figures

		Page
2.1	Ensemble Stacking	16
2.2	Three-way holdout split	16

Chapter 1

INTRODUCTION

Urban air monitoring needs to capture pollution changes in time and in many places. This helps find pollution problems and stop bad effects on health. Government monitoring stations are very accurate. They are expensive and can't be placed everywhere. Cost smart sensors can be used in more places but their readings can change because of temperature, humidity and the way the sensors get older. Some machine learning models like Random Forest and Support Vector Regression are used to fix these problems. These models can't predict extreme pollution levels, beyond what they have seen before. This study solves that problem by creating a system that uses sensors and a special way of combining different models. This new approach helps the system understand factors and still predict high pollution levels when needed.

1.1 Background of the Study

Ambient air pollution is known around the world as an environmental danger and a serious health problem. Being around pollutants like fine particles (PM2.5, PM10) and harmful gases such as nitrogen dioxide (NO2) ozone (O3) carbon monoxide (CO) and sulfur dioxide (SO2) leads to many early deaths each year and creates big financial problems globally [**concas**, **WHO1**]. Officials use fixed air quality monitoring stations with methods to check if rules are followed and to look at health risks [**giordano**, **WHO2**]. These proper tools are very accurate. Cost a lot to buy and keep running. This means there are not stations and each one covers a large area, sometimes several square kilometers [**concas**, **hayward**]. Because of that these systems miss areas where pollution is very high and also miss the small

changes in pollution levels that happen in busy cities [**concas**, **giordano**]

To fix the problem of not covering areas low-cost sensors have become a good choice. These sensors are cheaper and smaller making it possible to have more of them for coverage and to track personal exposure [**concas**]. Even though they can cover places they measure air directly and have big errors [**giordano**, **hayward**]. When measuring particles optical sensors guess the weight from how light bounces off them. This guess is wrong if the particles are not all the size or shape [**giordano**]. High humidity makes the particles bigger which tricks the sensors into thinking there is pollution than there really is [**giordano**, **hayward**]. When measuring gases the sensors use reactions that change with temperature. This causes them to mix up signals from gases especially between NO2 and O3 [**hayward**]. Also as the sensors age and are used more their readings change over time [**concas**, **hayward**].

Since the numbers from the sensors are not close to the values they must be checked against the proper instruments in the same place [**concas**, **hayward**]. Simple models like linear regression are often used because they are easy. They can't handle the complicated and changing relationships between the sensor data, the weather and other air chemicals [**hayward**]. Some advanced models like Random Forest, Support Vector Regression and Artificial Neural Networks have been used to better deal with these issues but each has its own problems [**concas**, **hayward**]. Random Forest can't predict when pollution is higher, than what it was trained on. It gives wrong numbers when pollution is very bad [**hayward**]. Support Vector Regression needs the settings and doesn't work well with time-based data [**concas**, **hayward**]. Artificial Neural Networks need a lot of data. Use too much power [**concas**, **hayward**]. Studies show that no one model works best for all types of pollution and different weather [**concas**, **hayward**].

1.2 Problem Statement

Although non linear models like Random Forest and Artificial Neural Networks are widely used to calibrate low cost air quality sensors, they struggle to extrapolate. Decision tree models like Random Forest cannot predict values outside their training limits because their predictions rely on averaging values at the leaf nodes. Similarly, Neural Networks focus on the most common baseline readings, causing them to under predict rare peak concentrations. Because these models cannot extrapolate beyond historical data, they flatten or cut off high concentration spikes, making them unreliable for real time health warnings during severe pollution events.

1.3 Objectives

To design and develop an air quality monitoring system with stacked ensemble to fix extrapolation issue. The specific objectives are as follows:

- To design and assemble the physical IoT air monitoring system (measuring PM, SO2, NO2, O3, CO, Temp, RH).
- To develop and train the Stacked ensemble Random Forest + SVR + MLR algorithm to correct environmental biases and extrapolate boundary limits.
- To evaluate the algorithmic accuracy using statistical metrics like RMSE, R2

The items here should be sub-systems that will be made to solve the problem stated above.

1.4 Purpose of the Study

The purpose of this study seeks to benefit the following

1.4.1 The Public

Empowers communities with accessible, localized, and real-time air quality data.

1.4.2 Government

Assists regulatory bodies like DENR in validating and planning low-cost sensor networks.

1.4.3 Academe

Provides robust hybrid ML calibration models for future environmental research.

1.4.4 ECE Field

Enhances physical hardware sensor node reliability through advanced drift correction.

1.4.5 SDG 3: Good Health

Supports public health by providing accurate, real-time exposure data to mitigate respiratory risks.

1.4.6 SDG 9: Innovation

Enhances localized sensor technology and develops robust machine learning calibration models.

1.4.7 SDG 11: Cities

Enables dense, scalable air quality monitoring networks to build sustainable urban communities.

1.4.8 SDG 13: Climate

Fosters environmental resilience through accessible and reliable localized air pollution data.

1.4.9 SDG 17: Partners

Connects academe, regulatory bodies (DENR), and local communities in collaborative research.

1.5 Definition of Terms

In the context of this project, the following are the operational definition of terms:

Calibration: the process of adjusting and verifying the accuracy of an instrument to ensure that it provides the correct readings within the specified tolerance levels Specify your references by placing citations.

Drift: Every physical sensor experience performance degradation over time due to continuous material aging, environmental exposure, and chemical depletion.

Ensemble: refers to the methods of combining multiple machine learning models

Overfitting: happens when a model learns the training data too well, memorizing its random noise and minor quirks instead of the real patterns

Underfitting: happens when a model learns the training data too well, memorizing its random noise and minor quirks instead of the real patterns

Chapter 2

THEORETICAL BACKGROUND

This chapter explores into a comprehensive review of theories and literature, including principles, concepts, and ideas applicable in the study. The first part of this chapter includes the discussion of the concept of Calibration theory, Statistical Process Control and Machine learning. The second part of this chapter includes the review of related literature. The Literature Review section synthesizes recent applied research in environmental monitoring. This discussion is organized into targeted subsections evaluating different machine learning methodologies ranging from standalone regression techniques to advanced stacked ensembles and their efficacy in mitigating non-linear cross-sensitivities and hardware drift. Ultimately, this chapter equips the reader with a comprehensive understanding of the physical vulnerabilities inherent in low-cost sensors and the data-driven algorithms necessary to correct them.

2.1 Theories

2.1.1 Calibration theory

Calibration process involves comparing the results of the device under test to a reference standard and making necessary adjustments to align the device's readings with the standard [**1**]. The foundational method used in all calibration is known as Zero and Span Adjustment (Two-point Calibration) which measure two known reference conditions typically a low point near zero and a high point near maximum range. This assumes a linear change in the measured device creating a slope in its error. The Slope for the adjustment is defined as [**2**]

$$M = \left(\frac{A_2 - A_1}{B_2 - B_1} \right)$$

Where:

- M = slope
- A = True Value
- B = measured value

2.1.2 Statistical Process Control

Statistical process control is used for controlling a process using statistical methods to detect the need for calibration [**4**]. For sensors it is critical to detect when the instrument has drifted. This can be done by applying different methods of statistical process control based on how the drift of the sensor behaves.

Standard deviation is a statistical method that measures how far numbers in a dataset are spread out from the mean [**7**]. The equation for standard deviation is defined as

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (a_i - \mu)^2}{n - 1}}$$

where:

- σ = Standard deviation
- a = place value holder of the entry in the dataset
- μ = mean
- n = number of entries in the dataset

Cumulative Sum (CUSUM) is a statistical process control method specifically designed to identify subtle, continuous sensor drift over extended deployment periods. Rather than evaluating daily readings in isolation, CUSUM tracks process stability during consistent baseline conditions and accumulates small, deviations that fall below the expected target mean [**5**]. When the accumulated penalty tally crosses a predefined decision limit typically set at 5σ of the baseline variance, the system mathematically confirms permanent hardware degradation and triggers a recalibration routine. The CUSUM equation is defined as [**6**]

$$S = \max(0, S_{t-1} + (\mu - E) - c)$$

where:

- S = current CUSUM
- S_{t-1} = previous CUSUM
- c = input
- E = error margin

A drift is detected when $S > 5\sigma$

2.1.3 Machine Learning

Machine learning is a subfield of artificial intelligence that uses algorithms trained on data sets to create models capable of performing tasks that would otherwise only be possible for humans [8]. A machine learning model receives a dataset D = X, Y

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

where:

- X = input parameters
- Y = target parameters
- x = input features
- y = target
- n = number of entries
- m = number of features

Random forest

Random forest is an ensemble of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest [**9**]. The generalization error for forests converges to a limit as the number of trees in the forest becomes large. The generalization error of a forest of tree classifiers depends on the strength of the individual trees in the forest and the correlation between them.

1. **Hyperparameter**

   Random Forest has three standard hyperparameters that are set to configure how the model behaves.

   - Trees (k) = Refers to the number of trees within the forest, more trees give a more accurate prediction at the cost of more computational power.

- Depth (p) = Defines how many times each node can split, bigger depth can increase prediction accuracy, but it can also lead to overfitting.
- max_Features (f) = Refers to the number of input features that are sampled per node.

2. **Bootstrapping**

   For each tree n number of entries are randomly selected where entries can be repeatedly selected such that $d_k \subseteq D$

   where:

   - d = randomly selected dataset
   - k = decision Tree

3. **Node Splitting**

   After each tree is assigned with a randomly selected dataset d a node is created to split the data into two classes, left(L) and right(R). To determine which data falls to L & R a threshold for every element of d is calculated based on the middle value of each element

$$s_x = \frac{\max(x) + \min(x)}{2}$$

   where:

   - s = threshold

   The values are segregated per features such that every value of the features that are more than the threshold are placed in R while the remaining are assigned to L

$$R = \{a \in x \mid a > s\}$$

$$L = \{a \in x \mid a \leq s\}$$

After all data for each element have been classed, the MSE of each is Calculated

$$MSE_x = \frac{|L_x|}{|x|} \sum_{i \in L_x} (L_x - \overline{L_x})^2 + \frac{|R_x|}{|x|} \sum_{i \in R_x} (R_x - \overline{R_x})^2$$

The node is split into L & R based on the element with the lowest MSE. Node Splitting will keep on recurring until p is reached. At the last node a leaf is calculated to represent the entire tree which is calculated as

$$\hat{Y}_N = \sum_{i \in Y} i$$

Where:

- $\hat{Y}$ = leaf
- N = Node

4. **Bagging**

   During testing the inputs are passed to each tree containing the calculated nodes. Every tree will have a leaf to represent their prediction. To get the final prediction for the model, bagging is calculated as

$$Y(x) = \sum_{i=1}^{k} \hat{Y}_i(X)$$

   where:

   - $Y(x)$ = RFR prediction

Support Vector Regression (SVR)

Support Vector Regression is a margin-based regression method that ignores small errors intentionally, it handles nonlinear relationships through kernels and holds up on noisy real-world data where standard regression comes up short [**10**].

1. **Standardization**

   All inputs are standardized by calculating the mean and the standard deviation of the dataset, calculated as

$$\mu_g = \frac{\sum_{i \in D_g} i}{|D_g|}$$

$$\sigma_g = \sqrt{\frac{\sum_{i \in D_g} (i - \mu_g)^2}{|D_g|}}$$

   where:

   - g = columns in the dataset

   Then standardized as

$$Z_g = \frac{D_g - \mu_g}{\sigma_g}$$

   where:

   - Z = standardized input

2. **Hyperparameters**

   Kernel ($K(x_i, x_j)$)- The kernel determines how SVR handles patterns in the dataset.

   Regularization (C) - C controls how much SVR penalizes errors on points outside the tube. High C means the model takes those errors seriously and

fits more tightly. Low C means the model accepts more violations in exchange for a simpler, flatter function.

Epsilon (ε) - Epsilon defines the width of the tolerance margin around the fitted function. A larger ε means a wider tube.

Gamma (γ) - γ is a special hyperparameter that only exist when the kernel is set to RBF. γ defines how far the influence of a single training example reaches, a lower γ treats distant points as similar, creating a smoother, more generalized, and linear-like curve.

3. **Kernel**

   The Radial Basis Function (RBF) kernel is used to train nonlinear SVR models which is defined as

$$K(x_i, x_j) = e^{-\gamma \|z_i - z_j\|^2}$$

   where:

   - i, j = indices for the dataset

4. **Weight**

   The weight of vectors are calculated as

   Where α_i and α_j^* are calculated as

$$\max_{\alpha_i, \alpha_i^*} \left[-\frac{1}{2} \sum_{i,j=1}^{n} (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) - \epsilon \sum_{i=1}^{n} (\alpha_i + \alpha_i^*) + \sum_{i=1}^{n} z_i (\alpha_i - \alpha_i^*) \right]$$

   where:

   - W_i = weighted vectors

- z = standardized target

With a constraint such that

$$\sum_{i=1}^{n}(\alpha_i - \alpha_i^*) = 0$$

$$0 \leq \alpha_i, \alpha_i^* \leq C$$

5. **Bias**

   A bias of the SVR is calculated as

$$b = z_I - \varepsilon - \sum_{i=1}^{n}(\alpha_i - \alpha_j^*)K(x_i, x_I)$$

   where:

   - b = bias
   - I = chosen index

6. **Testing**

   Using the calculated vectors in the training phase, the predicted values are calculated using the SVR general equation

$$f(x) = \sum_{i=1}^{n}(W_i K(x_i, x) + b)$$

   where:

   - $f(x)$ = SVR prediction

Multiple Linear Regression

Multiple regression is an ensemble of linear regressions which is a statistical method used for predictive analysis. It models the relationship between a depen-

dent variable and one or more independent variables by fitting a linear equation to the data. Multiple Linear Regression specifically extends this concept to include two or more independent variables [**13**]. In MLT beta is describes the linear relationships of the inputs which is defined as

$$\beta = (w^T w)^{-1} w^T Y$$

Beta is calculated using a matrix where:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix} \quad w = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1m} \\ 1 & x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}$$

Using the calculated β in the training phase the predicted value is calculated as

$$g(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_m$$

where:

- $g(x)$ = MLR prediction

2.1.4 Stack Ensemble

Stack Ensemble is a machine learning technique that builds a new model by combining the predictions of several previously trained models. Stack ensemble takes advantage of mutual complementarity among the base models to improve performance and enhance generalization ability [**11**]. Stack ensemble works by training multiple models as the basic learner and a second layer of model to serve as the meta learner. The meta learner combines the prediction of the basic learners to create its own prediction as shown in figure 2.1.3.

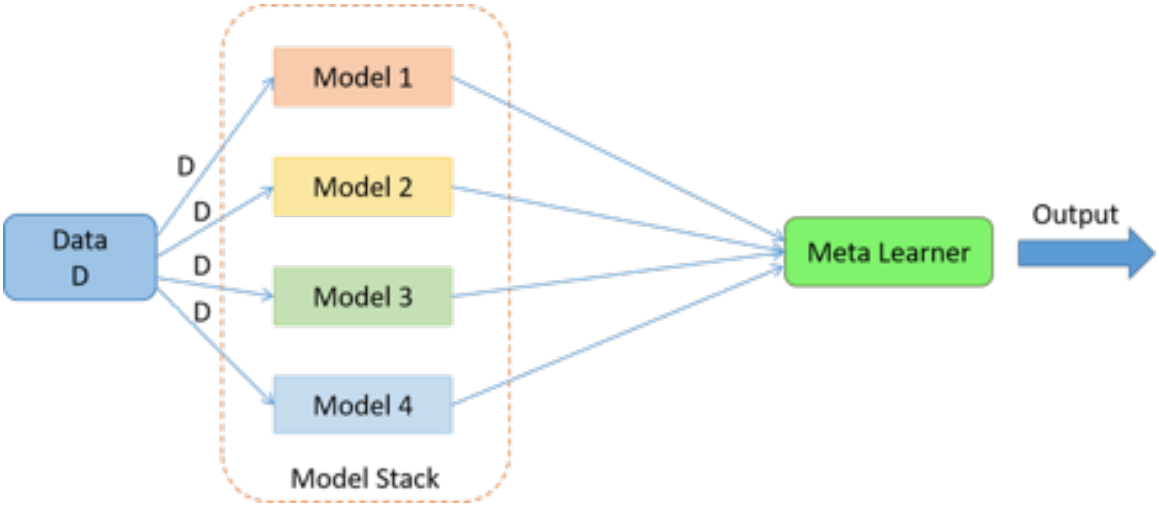


Figure 2.1: Ensemble Stacking

Three-way Holdout Split

Three-way holdout split is a method for stack ensemble to divide the dataset for training and validation [**12**]. For a dataset with less than 100,000 entries a split of 50% basic learner, 25% meta learner and 25% validation is a standard. However, for datasets with more than 100,000 entries a split of 98% basic learner, 1% meta learner and 1% validation is used as shown in figure 2.1.3.1.

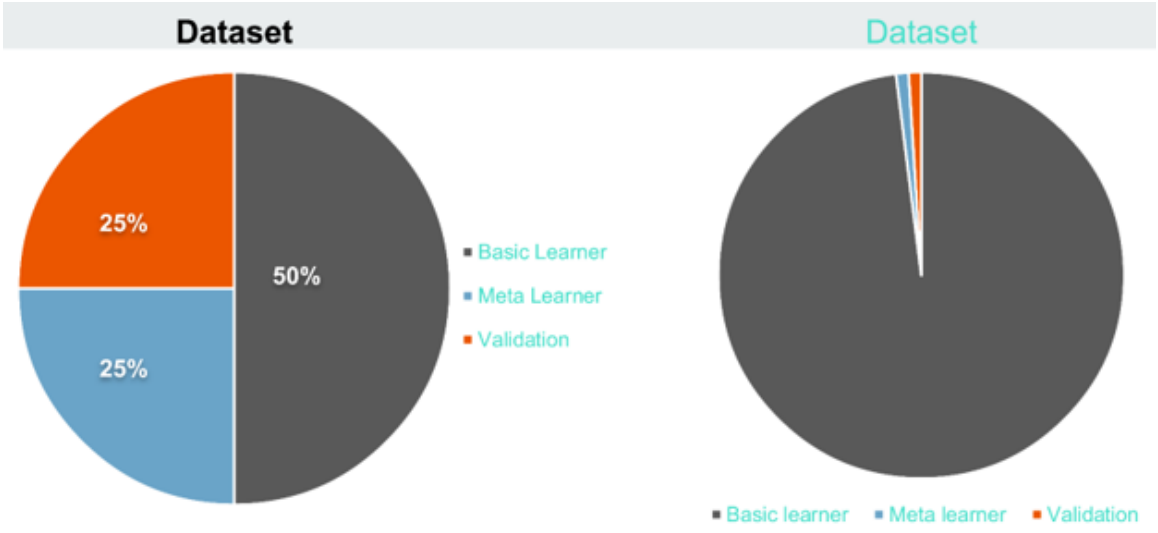


Figure 2.2: Three-way holdout split

2.1.5 Validation

Validating the machine learning model performance is a necessary step in machine learning to confirm that the model achieves its intended purpose. Validation is done right after a model is trained and exposed to unseen data [**14**, **17**]. Countless validation methods exist for machine learning, however the two methods which are considered as standards for regression models are RMSE and R^2.

1. **Root Mean Square Error**

   The root mean square error (RMSE) has been used as a standard statistical metric to measure model performance [15]. RMSE measures the average amount of error per prediction, meaning it is in the same unit as the output. RMSE is calculated as

   $$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - h_i)^2}$$

   where:

   - h = predicted value of the datapoint

2. **Coefficient of Determination**

   Coefficient of Determination (R^2) is used to measure how well a model fits data and how well it can predict. R^2 determines how much the variation is in the data. The closer the R^2 value is to 1, the better the model fits in the data [16]. The R^2 is calculated as

   $$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - h_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

   where:

   - $\bar{y}$ = mean of target values

2.2 Literature Review

This section investigates the application of machine learning (ML) algorithms for the in-field calibration of low-cost air quality sensors monitoring particulate matter (PM), sulfur dioxide (SO2), nitrogen dioxide (NO2), ozone (O3), and carbon monoxide (CO). While low-cost sensors present a scalable solution to achieve high spatial and temporal resolution in air quality monitoring, their raw measurements often exhibit poor correspondence with professional-grade reference stations. The reliability of these devices is heavily compromised by internal hardware errors and external environmental interferences. Specifically, their performance is notoriously sensitive to non-linear factors such as temperature variations, relative humidity, cross-gas sensitivities, and progressive sensor drift. To bridge this accuracy gap, recent research highlights machine learning as an essential calibration tool. By effectively modelling the complex, non-linear dependencies between environmental confounders and target pollutant signals, ML techniques can dynamically correct sensor data and significantly enhance overall measurement reliability.

2.2.1 Comparing different calibration method

Various studies have evaluated the efficacy of different machine learning (ML) algorithms in calibrating low-cost environmental sensors. For example, [**tastan**] assessed the performance of multiple ML models in calibrating a low-cost PMS7003 particulate matter (PM2.5) sensor. By integrating temperature and humidity data from an AHT10 sensor and referencing a Dienmern DM72b monitor, the study demonstrated high predictive accuracy across all tested algorithms. Specifically, tree-based ensemble methods such as AdaBoost (R2 = 0.991, RMSE = 2.541) and Random Forest (R2 = 0.987, RMSE = 2.212) slightly outperformed traditional approaches like Support Vector Machines (R2 = 0.977, RMSE = 2.647). The

researchers concluded that integrating meteorological variables significantly improves the accuracy of PM2.5 predictions.

Similarly, [**wang**] investigated the calibration of Alphasense sensors for both PM2.5 and nitrogen dioxide (NO2) against Teledyne T640 and Thermo Environmental 42C reference instruments. The authors tested a suite of algorithms, including Random Forest, Support Vector Regression, Artificial Neural Networks, and LightGBM. They found that the Generalized Additive Model exhibited the weakest performance (lowest R^2 and highest RMSE). The researchers attributed this lower accuracy to the mobile nature of the deployment and the relatively low pollution levels in the tested areas.

Expanding on the application of ML across broad geographic areas, [**yaqoob**] utilized PurpleAir PM2.5 sensors referenced against federal state monitors to evaluate calibration models across multiple states. The researchers compared Random Forest, Linear Regression, Gradient Boosting Regressor, K-Nearest Neighbors, and Neural Networks over a four-year period. Their results indicated that Neural Networks yielded the highest predictive accuracy (R2 = 0.88, RMSE = 1.75 μg/m^3), while traditional Linear Regression and K-Nearest Neighbors demonstrated the weakest performance (both achieving an R2 = 0.83). Notably, the study found that simply adding temperature as a linear correction factor did not significantly improve model performance; instead, non-linear ML models were required to successfully capture the complex interdependencies between environmental features across widespread datasets.

Recent research has also explored the calibration of agricultural sensors over extended temporal periods. [**killeen**] investigated nitrous oxide (N2O), pH, and temperature measurements using an LI-8250 Multiplexer and LI-780 Trace Gas Analyzer, referenced against automated gas chambers over a four-year period. The study compared traditional ML models (Random Forest, Linear Regression, Multi-

layer Perceptron, Support Vector Machine) with tensor-based time-series models (1D-CNN, LSTM). The authors found that LSTM emerged as the most robust and consistent model for both interpolation and multi-year extrapolation tasks (achieving an R2 of 3.30 and RMSE of 3.30 for short-term predictions). Crucially, these non-linear models successfully captured complex agro-environmental interactions without the need for site-specific, process-based parameters.

However, the long-term stability of these calibration models remains a significant challenge. [**barrett**] evaluated the calibration of a Winsen MH-Z19C NDIR CO2 sensor against a Picarro G2401 gas analyzer using models such as Random Forest Regression (RFR), Support Vector Regression (SVR), 1D-CNN, and a hybrid CNN-LSTM. Utilizing holdout validation and forward-in-time cross-dataset testing, the study revealed that while calibration accuracy was maintained over a two-month gap between training and testing, performance degraded severely when extrapolated over a 7-month unseen dataset (with all architectures exhibiting weak to negative correlation). Ultimately, the authors found that SVR was the best overall performing model due to its ability to preserve the statistical distribution profile of the reference data while maintaining low error rates. Furthermore, the study noted that training on shorter, contiguous time segments generated lower prediction errors than training on broader time spans, highlighting a major limitation of highly computational, data-heavy models like Neural Networks.

2.2.2 Calibration using random forest

The Random Forest (RF) algorithm is widely utilized in air quality monitoring due to its robust ability to reduce model overfitting [**concas**]. Several studies have demonstrated its efficacy in calibrating low-cost sensors when they are co-located with reference stations. For instance, [**Tiago**] successfully mitigated cross-sensitivities for CO2 measurements by integrating relative humidity and tempera-

ture data, achieving an R2 of 0.93 and an RMSE of 10.93. Their work highlights that in RF training, data diversity is often more critical than dataset size. Similarly, [**Ionascu2021**] utilized RF to manage cross-sensitivities in Alphasense gas sensors, reporting high correlations for CO (R2 = 0.98) and O3 (R2 = 0.92), although they noted that the sensors struggled at lower pollution detection limits. Supporting these findings, Bush et al. (2022) calibrated multi-pollutant sensors (PM2.5, PM10, and NO2) against reference stations, demonstrating that RF models offer multivariable efficiency, require minimal preprocessing, and are robust to collinearity, yielding strong R2 values ranging from 0.79 to 0.91.

Conversely, other studies have highlighted the challenges of applying Random Forest in less ideal environmental conditions. While their earlier work was highly successful, a subsequent study by [**ionascu2024**] demonstrated that RF models struggle when gas concentrations fall near the sensor's lower detection limits. In measuring SO2, NO, and particulate matter (PM) using Alphasense and Plantower sensors, they achieved strong correlations for PM (R2 = 0.94). However, the models yielded significantly lower accuracy for gases like NO (R2 = 0.20) and SO2 (R2 = 0.61) due to the low ambient concentrations in the co-located testing areas. Furthermore, [**han**] illustrated the critical importance of calibration protocols for RF training. In their measurement of CO, NO2, O3, and SO2, the absence of direct co-location with a reference station resulted in notably poor model performance, yielding low R2 values (e.g., SO2 = 0.012, NO2 = 0.45) and high RMSE readings. Together, these studies suggest that while RF is a powerful calibration tool, its efficacy is highly dependent on sufficient pollution levels and strict co-location procedures.

Despite these successful implementations, Random Forest models exhibit significant limitations in long-term field deployments. A primary constraint is their inability to extrapolate readings beyond the numerical boundaries of their training

data, which can lead to severe inaccuracies during unexpected pollution spikes. Furthermore, RF models struggle with generalizability; their predictive accuracy often degrades when seasonal weather patterns shift outside the parameters observed during their initial training phase. Finally, these models face inherent scalability challenges; because they are heavily reliant on the specific environmental conditions of their co-location site, transferring a trained RF model to a new geographical location typically requires a complete recalibration.

2.2.3 Stacked ensemble and hybrid approaches

Several studies have explored the efficacy of stacked, ensemble, and hybrid machine learning algorithms across various predictive modeling domains. For instance, [**Weimin Wu**] investigated the use of a stacked ensemble machine learning model to predict the n-octanol/air partition coefficient (KOA) of Polybrominated Diphenyl Ethers (PBDEs). Their study utilized a super learner framework that integrated Random Forest (RF) and Support Vector Regression (SVR) as base learners in the first layer, with Multiple Linear Regression (MLR) functioning as the meta-learner in the second layer. By leveraging existing chemical databases and density functional theory (DFT) simulations, the researchers demonstrated that this hybrid approach reduced extrapolation errors by 32% to 48% compared to individual algorithms. The resulting RF-SVR ensemble model achieved high predictive accuracy, yielding an R2 of 0.981 and a Root Mean Square Error (RMSE) of 0.230.

Similarly, [**tang**] applied a hybrid modeling approach to estimate high-resolution PM2.5 mass concentrations over large geographical areas. By combining satellite remote sensing data with observations from meteorological stations, they developed a hybrid model integrating RF and Extreme Gradient Boosting (XGBoost). This RF-XGBoost ensemble yielded an improved RMSE of 7.25 µg/m3, outperforming a standalone RF model, which produced an RMSE of 9.51 µg/m3. This

highlights the effectiveness of combining robust tree-based models to capture complex spatial and environmental variables for reliable air quality prediction.

Recent research has also extended these hybrid methodologies to real-time state-of-charge (SOC) estimation in battery management systems (BMS). [**tau**] developed a Hybrid Ensemble Dual-State Kalman Filter (HEAD-KF) deployed on an edge computing node (Raspberry Pi 4 Model B). The experimental setup utilized a 20S1P lithium-ion pack with NCA chemistry cells, managed by a BMS with an analog front-end, and evaluated using a 3 kW electronic load cycler. The estimation framework fused XGBoost and RF regressors through non-negative ridge stacking. The output of this ensemble was then refined by a dual-state Kalman Filter, where the process and measurement noise covariances were adaptively auto-tuned in real-time based on residual variance analysis. This methodology perfectly bridges the gap between accuracy and computational cost by combining lightweight, data-driven machine learning with a physics-based, statistical smoother. The approach effectively rejected both high-frequency measurement noise and low-frequency sensor drift, demonstrating the viability of hybrid edge-AI solutions for precise embedded battery management.

Chapter 3

RESEARCH METHODOLOGY

Describe the organization or flow of the contents in this chapter before the start of the first section. What can your readers learn from this chapter? How did you organize your discussions and subsections? You can add subsections but NOT sections.

Description of Subjects can be removed if this is not applicable to your research.

3.1 Proposed Method

Aside from the discussion regarding the proposed method, this must contain the block diagram(s) and possible flowchart(s). Note that supporting circuits, block diagrams and flowcharts should be in the appendix. For components, the order should be INPUT, PROCESS, OUTPUT. The sequence should be as follows: system block diagram, flowchart(s), components, circuit(s), and final project.

DO NOT FORGET: All figures and tables should be mentioned in the paragraph or discussion. Discuss the system block diagram in detail. Each block should contain the process NOT the result of the process. Variable names and the flow shown in the flowcharts should also be explained. For components, the following should be included: picture of actual component; model number; technical specifications (only those that are significant for the project); purpose or function in the prototype; and reason for selection. For the proposal hearing, CAD model with dimensions of the prototype should be shown and discussed. After the final defense, include the actual picture at different angles and explain its step-by-step operation.

3.2 Description of Experiments

During the proposal period, describe in detail the possible experiments that will be performed. For each experiment, it must have the objective(s), materials, and procedure. Each subsection here should be an experiment. All the details, settings, and assumptions should be discussed. Tables may also be presented here.

3.2.1 Title of Experiment of 1

This experiment aims to measure something. Note that each experiment should have a measureable objective that will help you determine/test a specific aspect/performance of your prototype. You should VARY ONLY ONE VARIABLE PER EXPERIMENT. The materials required are:

1. material 1;
2. material 2; and
3. specify program name and include this in the appendix.

The procedure is as follows:

1. Each item here should be something to be done.
2. Descriptions or expected changes can be included RIGHT AFTER a specific procedure.
3. For those that require tables, show the template of the table and specify in the procedure the column where the data should be inputted.
4. If a graph will be required, spcify the type of graph and data to be used.
5. Lastly, instructions on processing the collected data should also be mentioned here.

3.2.2 Title of Experiment of 2

This experiment aims to measure something. The materials required are:

1. material 1
2. material 2

The procedure is as follows:

1. Load the program to something.
2. Place one part in a corner.

3.3 Description of Subjects

This section is optional or may be changed depending on your project. For those who will use datasets, you can discuss the details here.

Chapter 4

RESULTS AND DISCUSSIONS

Describe the organization or flow of the contents in this chapter before the start of the first section. What can your readers learn from this chapter? How did you organize your discussions and sections? Each section in this chapter will be your experiment, which were discussed in the previous chapter. The title of the experiment in the previous chapter should also be the same here. Discuss, in detail, your data and interpretation of the results since these will be used as basis of your conclusion and recommendation.

4.1 Title of Experiment of 1

Show your tables, plots, and figures here. Discuss the contents in detail and the meaning of the data. For tables and plots, discuss first what is shown then its meaning and/or significance. Any problems encountered in the experiment should also be mentioned here.

4.2 Title of Experiment of 2

Show your tables, plots, and figures here. Discuss the contents in detail and the meaning of the data. The number of experiments described in Chapter 3 should be the same here. Positions of tables and figures SHOULD BE FIXED when finalizing the document in PROJECT 2 or AFTER THE FINAL DEFENSE.

NOTE: The number of subsections or experiments in Section 3.2 should be equal to the number of sections in this chapter.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

Describe the organization or flow of the contents in this chapter before the start of the first section. What can your readers learn from this chapter? How did you organize your discussions sections? Section names should not be changed and subsections should not be added. Cross-references are also prohibited here. Your statements must be generalizations.

5.1 Conclusions

The first sentence should state what this thesis has accomplished. Make sure to connect or relate your objectives with your proposed method and the collected data. Statements should be based on the data, findings, and observations in the experiments. Conclusions that have no basis in previous chapters are **not acceptable**.

5.2 Recommendations

This section can include recommendations on how to improve your method. Improvement refers to technical aspects of the research and should not be about merely changing components or its aesthetic aspects. You can also discuss the problems or issues encountered or future works related to your research topic. Others identify the inherent limitations of their method then discuss how to overcome them.

NOTE: Before a recommendation, explain first the situation or its basis.

BIBLIOGRAPHY

Appendix A

PRIOR ART SEARCH and ANALYSIS REPORT

Innovation and Technology Support Office
Cebu Institute of Technology - University
ITSO_IPPF002av01

ABSTRACT and KEYWORD TABLE

Name	
Program and Yr.	
Working Title	

Abstract
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

Keyword Table

SS01	SS02	SS03	SS04	SS05	SS06	SS07	SS08	SS08	SS10

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ABSTRACT and KEYWORD TABLE

Name	
Program and Yr.	
Working Title	

Abstract
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

Keyword Table

SS01	SS02	SS03	SS04	SS05	SS06	SS07	SS08	SS08	SS10

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IPC SEARCH REPORT

TITLE

SEARCH STRING	IPC	IPC DESCRIPTION	REMARKS
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
			☐ utilize ☐ disregard
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			☐ utilize ☐ disregard
			☐ utilize ☐ disregard

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CERTIFICATION

SEARCH CONDUCTED BY	SEARCH REVIEWED BY
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:
Institutional Research Code:	

PATENT SEARCH REPORT

TITLE	FILING DATE

ABSTRACT
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention

CLASSIFICATION	KEYWORDS

CONFIDENTIAL

SEARCH STRINGS AND DATABASES

DATABASE	SEARCH STRING	NUMBER OF HITS

RELEVANT DOCUMENTS

CATEGORY	PATENT NO	TITLE/ABSTRACT	PUBLICATION DATE	RELEVANT TO CLAIM

* **Special Categories of Cited Documents:**

"A" document defining the general state of the art which is not considered to be of particular relevance

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

Note: Further references not indicated in this search report may be cited during substantive examination.

NON PATENT LITERATURE

CATEGORY	REFERENCE	RELEVANT TO CLAIM

PRIOR ART SUMMARY

PATENT NO. / REFERENCE TITLE	SIMILARITIES	DISSIMILARITIES	UNIQUE FEATURE

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CERTIFICATION

SEARCH CONDUCTED BY	SEARCH REVIEWED BY
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:
Institutional Research Code:	

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Appendix A

PATENT DRAFT

SUBJECT MATTER ELIGIBILITY DETERMINATION FOR ICT AND CII

TITLE
ABSTRACT
What does it do? Essential function of invention What is it made of? Physical structure of the invention What is the end result? Essential effect or basic product resulting from the invention What is it used for? Intended use for the invention
PRIOR ART
STEP 1: Is the claim involving the field of ICT/CII?
STEP 2: Is the claim directed to a product or process within the meaning of the IP Code?
STEP 3: Does the claim involved an abstract idea?
STEP 4: Does the claim constitutes technical character wherein a technical solution to a problem is evident after weighing all the factors?

SEARCH CONDUCTED BY	
Signature and Date:	Signature and Date:
Complete Name:	Complete Name:
Department:	Department:

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INVENTION SPOTTING: INVENTIVE STEP SPOTTING

Claim	Your Invention:		CPA1: (number)	CPA2: (number)	CPA3: (number)
	(Title)		(Title)	(Title)	(Title)
TF1: TE:					
TF2: TE:					
TF3: TE:					
TF4: TE:					
TF5: TE:					
TP:					
TS:					

TF - Technical Feature TE - Technical Effect TP - Technical Problem TS - Technical Solution

A. 1. Find Novelty CPA (Y / N) 2. TFD - Technical Feature Difference (Y / N) 3. TED - TE Difference (Y / N) 4. Obvious? (Y / N)
5. Decision:

B. Modify TP? (Y / N):
Unexpected TE?:

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[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 1

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APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 2

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APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 3

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APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 4

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APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

[insert figure, if more than one can fit in this page, merge this with the next]

Fig. 5

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APPLICANT'S NAME
Applicant

By:

YOUR NAME AND SIGNATURE
Agent

(TITLE)

(short and specific, if possible, not more than 10 words)

TECHNICAL FIELD OF THE INVENTION

(Statement of the field of the art to which the invention pertains, should be directed to the subject matter, starts with "BROAD" statement [broad field of technology] e.g. Computers, Electronics & Communications, and ends with "SPECIFIC" statement [specific field or goal] e.g. improving light sensor detection, usually one sentence only.

BACKGROUND OF THE INVENTION

(Cite Closest Prior Art Reference [CPAR] and state technical problems with [CPAR] that the invention wants to solve)

SUMMARY OF THE INVENTION

(State unexpected advantages of the invention over the CPAR)

BRIEF DESCRIPTION OF THE DRAWINGS

(Must show every feature of the invention, figures must be consecutively numbered, the elements must be designated by way of reference numerals corresponding to the elements in the description)

e.g. Figure 1 shows a top plan view of the invention.

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Figure 2 provides a block diagram of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(State in detail the different parts or elements of the invention, how they interact, how the best embodiment can be manufactured or fabricated or replicated, and how the invention is used.)

(Broadening Clause)

The preferred embodiment of this invention is described in the above-mentioned detailed description. It is understood that those skilled in the art may conceive modifications and/or variations to the embodiment shown and described therein. Any such modifications or variations that fall within the purview of this description are intended to be included therein as well. Unless specifically noted, it is the intention of the inventors that the words and phrases in the specification and claims be given the ordinary and accustomed meanings to those of ordinary skill in the applicable art. The foregoing description of a preferred embodiment and best mode of the invention known to the applicant at the time of filing the application has been presented and is intended for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present invention to the precise form disclosed, and many modifications and variations are possible in the light of the above teachings.

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CLAIMS (separate sheet)

1.
2.
3.
4.
5.

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(TITLE)

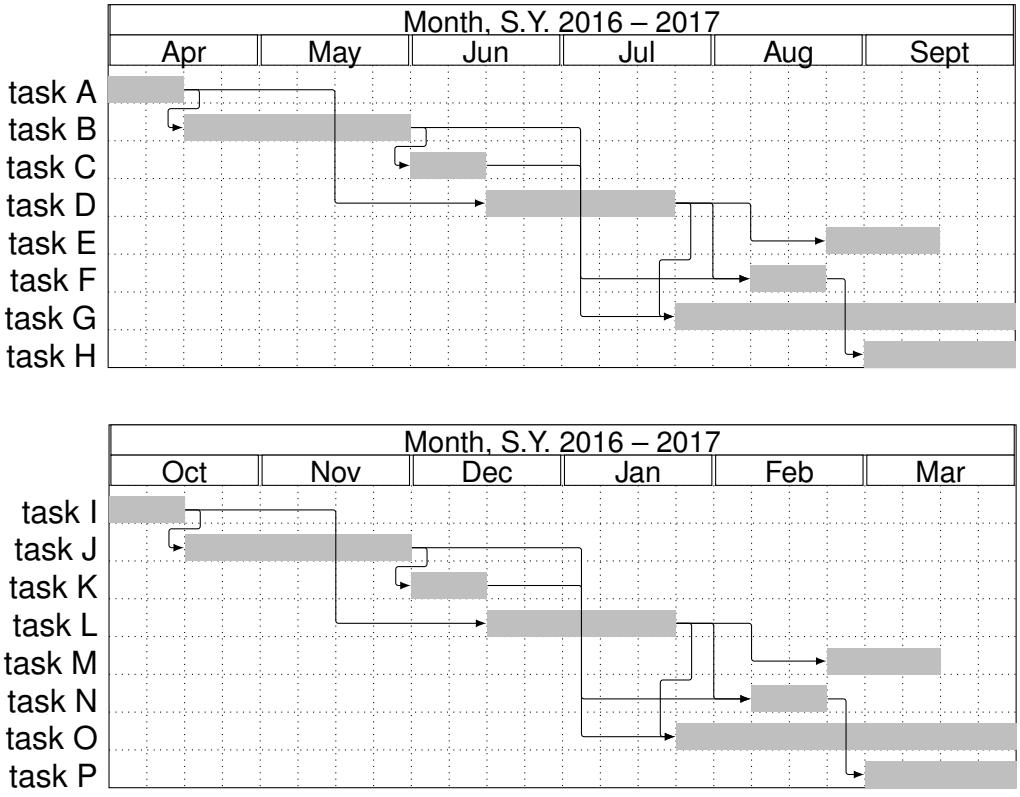
ABSTRACT OF THE DISCLOSURE (separate sheet)

(State the technical problem that your invention solves and the elements/structures/parts of your invention, in 150 words)

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Appendix B

TIMETABLE OF ACTIVITIES



Month, S.Y. 2016 – 2017
Apr
May
Jun
Jul
Aug
Sept
Oct
Nov
Dec
Jan
Feb
Mar
task A
task B
task C
task D
task E
task F
task G
task H
task I
task J
task K
task L
task M
task N
task O
task P
task P
task Q
task R
task S
task T
task U

Task	Task Description	FamilyN1	FamilyN2	FamilyN3	FamilyN4
A	Specific activity				
B	Specific activity				
C	Specific activity				
D	Specific activity				
E	Specific activity				
F	Specific activity				
G	Specific activity				
H	Specific activity				
I	Specific activity				
J	Specific activity				
K	Specific activity				
L	Specific activity				
M	Specific activity				
N	Specific activity				
O	Specific activity				
P	Specific activity				
Q	Specific activity				
R	Specific activity				
S	Specific activity				
T	Specific activity				
U	Specific activity				

Appendix C

RESEARCH BUDGET

This is the budget of the thesis or project. During proposal, tabulate the possible expenses here. You can use a **long table** if your list of expenses spans several pages. In order to prevent your tables from floating, use the **tabular** environment only. List of expenses can be shorter by grouping them according to semester or a certain span of time. Item description and quantity should be left justified and centered, respectively. Amount should be right justified with two decimal places and comma for those that reach or exceed 1,000.00.

Appendix D

ADDITIONAL FLOWCHARTS

This appendix is for supplemental flowcharts only. Those that are for the main program or system should be inside Chapter 3. Figures here should also be cross-referenced or mentioned in other chapters.

Appendix E

PSEUDO-CODES

Insert pseudo-code of your algorithms here. Use the `algorithmic` environment. Below is an example:

Algorithm 1 Euclid's algorithm

```
1: procedure EUCLID(a, b)                    ▷ The g.c.d. of a and b
2:     r ← a mod b
3:     while r ≠ 0 do                        ▷ We have the answer if r is 0
4:         a ← b
5:         b ← r
6:         r ← a mod b
7:     end while
8:     return b                              ▷ The gcd is b
9: end procedure
```

If additional programs used in the experiment will be inserted, CHANGE the chapter title in the main tex file (line 173) to Experiment Codes. First, indicate the name of the program/code with file extension (ex. test1.sci) then enclose the commands with the verbatim environment.

Appendix F

ELECTRICAL CIRCUITS

Insert supplemental circuit diagrams here. Examples are power supply circuits (customized for the prototype) and driver circuits (not part of the main methodology). Insert the diagram as **figure** and cross-reference or mention these in other chapters.

Appendix G

SUPPLEMENTAL DATA

Insert your raw data here in tabular form (data from 100 trials). The tables MUST HAVE numbers and captions, i.e. use the **table** environment. If applicable, use long table with headers and footers. You can also cross-reference the tables here in your discussions in other chapters. Note that the data here should be summarized in a separate table or graph that is placed in Chapter 4.

Appendix H

PLAGIARISM CHECK REPORT

Final Doc

ORIGINALITY REPORT

9%	8%	4%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	en.wikipedia.org Internet Source	1%
2	ipindia.gov.in Internet Source	1%
3	patents.google.com Internet Source	1%
4	T.B. Asafa, T.M. Afonja, E.A. Olaniyan, H.O. Alade. "Development of a vacuum cleaner robot", Alexandria Engineering Journal, 2018 Publication	<1%
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36	psasir.upm.edu.my Internet Source	<1%
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controlled robot: A simple approach", 2016 International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT), 2016
Publication

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	On		

Appendix I

ETHICS CLEARANCE

CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY
Institutional Ethics Review Committee
ethics@cit.edu

Month DD, YYYY

COMPLETE NAME OF PROPONENT/GROUP LEADER
Designation
Institution
Address

RE: **Title of project/study**
IERC Code: **Code**

Subject: Research Ethical Clearance

Dear **Name of Proponent/Group Leader**:

This is to acknowledge receipt of your request and the following supporting documents dated Month DD, YYYY.

1. Submitted document 1
2. Submitted document 2
3. Submitted document 3
4. Submitted document 4
5. Submitted document 5

The above documents underwent FULL/EXPEDITED review (for FULL:) during the meeting on Month DD, YYYY at Venue of Meeting.

The study protocol is hereby granted APPROVAL FOR IMPLEMENTATION by CIT-U IERC. This ethical clearance is valid for one year until Month DD, YYYY.

While the study is in progress, the Proponent/s is/are requested to submit the following documents:

1. Progress report using form IERC-PF-19: Application for Continuing Review every six months from date of approval including the following: (*NOTE: In view of the active ethical clearance, this report is mandatory even if the study has not started or is still awaiting release of funds.*)

    a. Date covered by the report
    b. Protocol summary and status report on the progress of the research

c. Number of participants accrued
d. Withdrawal or termination of participants
e. Complaints on the research since the last CIT-U IERC review
f. Any relevant multi-center research reports
g. Any relevant information especially about risks associated with the research
h. A copy of the informed consent document

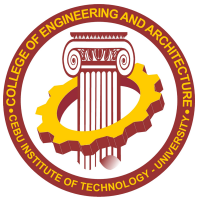
2. Any changes in the protocol, especially those that may adversely affffect the safety of the participants during the conduct of the study, including changes in personnel, must be submitted or reported using form IERC-PF-15: Application for Ethics Review of Amendments.
3. Revisions in the informed consent form using form IERC-PF-15: Application for Ethics Review of Amendments.
4. Reports of adverse events including from other study sites using the form IERC-PF-18: Reportable Negative Event Report.
5. Notice of early termination of the study and reasons for such using form ERC-PF-21: Early Termination Report.
6. Any event which may have ethical significance.
7. Any information which is needed by the CIT-U IERC to do ongoing review
8. Notice of time of completion of the study using IERC-PF-22: Final Report.
9. Application for renewal of ethical clearance 90 days before the expiration date of this approval through submission of form ERC-PF-19: Application for Continuing Review.

Very truly yours,

CHAIR's COMPLETE NAME
Chair

Appendix J

CURRICULUM VITAE



CONTACT INFORMATION

Name: Given Name MI. Family Name

Address: Number and Street, City or Province, Philippines, Zip Code

Telephone Number: 063 - area code - number

Cell Phone Number: 09numbers

Email: address@site.dmn

PERSONAL INFORMATION

Date of Birth: Month DD, YYYY
Place of Birth: City, Country
Citizenship: Filipino

EDUCATIONAL BACKGROUND

College
Cebu Institute of Technology University
N. Bacalso Avenue, Cebu City, Philippines
Bachelor of Science in Electronics Engineering
March 2013

High School
Name of High School attended
Address
Month YYYY

WORK EXPERIENCE

Name of Company (latest)
Address
Position
Responsibilities
Inclusive Dates

Name of Company
Address
Position
Responsibilities
Inclusive Dates

PROFESSIONAL QUALIFICATIONS

Certifications
certification 1 (latest)
certification 2

Computer Skills
skill 1
skill 2

HONORS AND AWARDS

Name of Award 1 (latest)
Institution
Month, YYYY

Name of Award 2
Institution
Month, YYYY

PROFESSIONAL AFFILIATION

Name of Organization 1 (latest)
Position
Inclusive Years

Name of Organization 2
Position
Inclusive Years

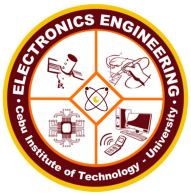
SKILLS AND INTERESTS

Skills: skill 1, skill 2, skill 3

Software: software 1, software 2, software 3, software 4, software 5

Community Extension: extension 1, extension 2

Languages: Language 1 - Native, Language 2 - Fluent, Language 3 - Advanced or Intermediate or Beginner



CONTACT INFORMATION

Name: Given Name MI. Family Name

Address: Number and Street, City or Province, Philippines, Zip Code

Telephone Number: 063 - area code - number

Cell Phone Number: 09numbers

Email: address@site.dmn

PERSONAL INFORMATION

Date of Birth: Month DD, YYYY
Place of Birth: City, Country
Citizenship: Filipino

EDUCATIONAL BACKGROUND

College
Cebu Institute of Technology University
N. Bacalso Avenue, Cebu City, Philippines
Bachelor of Science in Electronics Engineering
March 2013

High School
Name of High School attended
Address
Month YYYY

WORK EXPERIENCE

Name of Company (latest)
Address
Position
Responsibilities
Inclusive Dates

Name of Company
Address
Position
Responsibilities
Inclusive Dates

PROFESSIONAL QUALIFICATIONS

Certifications
certification 1 (latest)
certification 2

Computer Skills
skill 1
skill 2

HONORS AND AWARDS

Name of Award 1 (latest)
Institution
Month, YYYY

Name of Award 2
Institution
Month, YYYY

PROFESSIONAL AFFILIATION

Name of Organization 1 (latest)
Position
Inclusive Years

Name of Organization 2
Position
Inclusive Years

SKILLS AND INTERESTS

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